When Safety Judges Disagree: Systematic Stress Testing of LLM-as-Judge Evaluations

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Abstract

LLM-as-Judge has become a widely adopted paradigm for safety evaluation, yet existing evaluations implicitly assume that different judge models produce consistent assessments under identical criteria. In this work, we identify a critical reliability issue in LLM-based safety evaluation: judge disagreement can be systematically induced through adversarially discovered responses. We introduce TVD-Conflict, a Task-Validator-Data based agentic framework that treats multiple LLM judges as competing validators and searches for responses that cause conflicting safety judgments. Using TVD-Conflict, we construct a collection of disagreement-inducing cases across multiple safety evaluation scenarios and reveal systematic failure modes in current LLM safety judges, including sensitivity to implicit harmful intent, code-mediated behaviors, and multi-step contextual reasoning. Our findings demonstrate that the reliability of LLM-as-Judge evaluations can depend substantially on the choice of evaluator, highlighting the need to stress-test judges themselves before using them as trusted evaluation protocols.

1 Related Work

Reliability of LLM-as-a-Judge. Prior work has shown that model-based evaluators are susceptible to systematic biases. Ye et al. [2025] identify 12 types of bias in LLM-as-a-Judge and introduce CALM, a framework that uses automated, principle-guided modifications to quantify judge sensitivity across models. Their study evaluates predefined bias categories and consistency under controlled perturbations. Our work is complementary: rather than measuring a fixed set of biases within individual judges, TVD-Conflict searches for responses that induce conflicting safety decisions across multiple judges operating under the same evaluation criteria.

References

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